The II-XX Model

II-XX Model

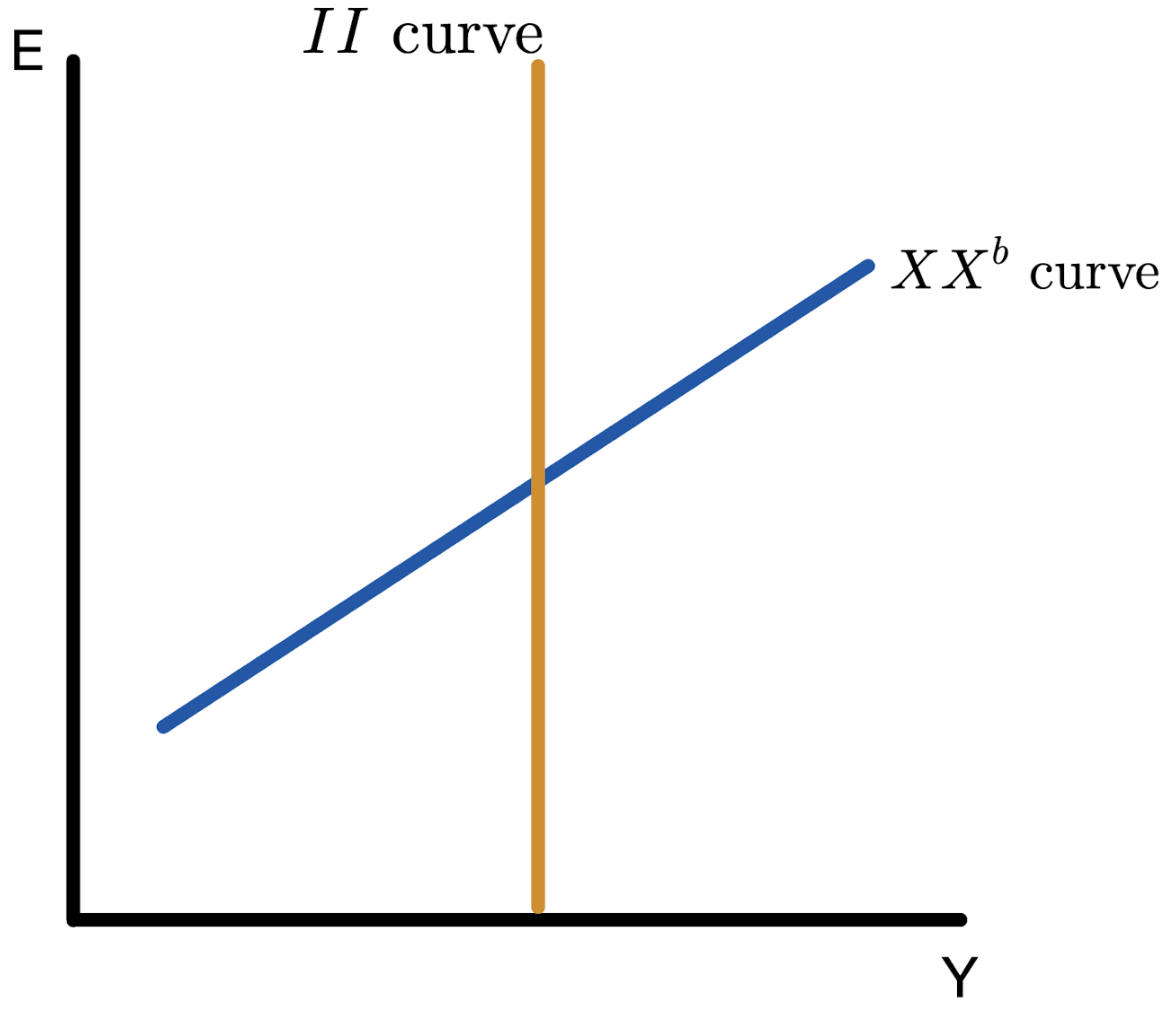
- Start with minimal assumptions
  - There are numbers Y^f and XX^b such the economy tends to $Y = Y^f$ and $CA = XX^b$ in the long run
  - $Y = Y^f$ and $CA = XX^b$ are desirable economically or politically
  - Monetary and fiscal policy cannot influence Y^f and XX^b, but can affect how long it takes for Y and CA to get to Y^f and XX^b

II-XX Model

- Do not assume
  - Equilibria in any markets
  - Mechanisms for how Y and CA approach Y^f and XX^b

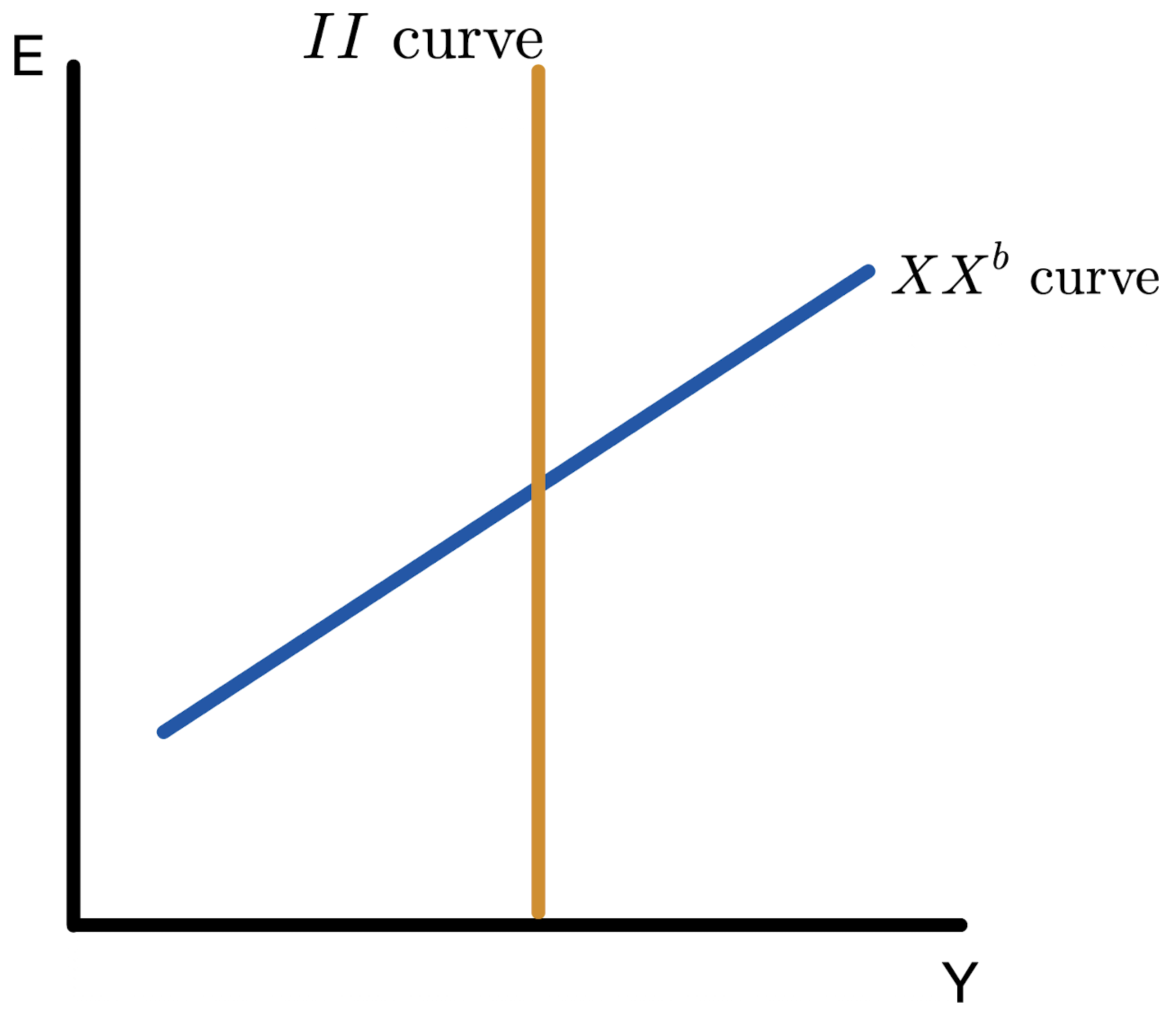
Internal and External Balance

- II curve: (Y, E) consistent with **internal balance** $Y = Y^f$

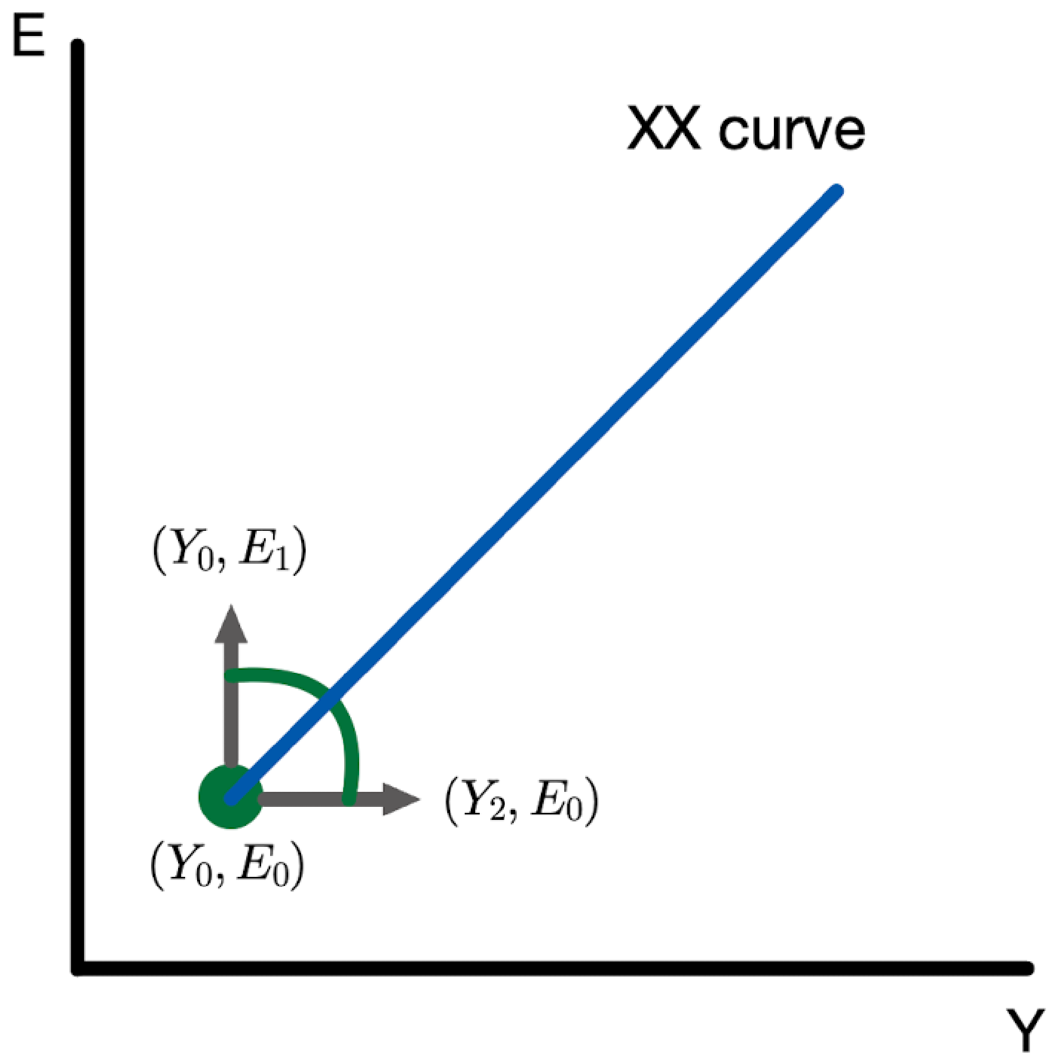


Internal and External Balance

- II curve: (Y, E) consistent with **internal balance** $Y = Y^f$
- XX curve: (Y, E) consistent with **external balance** $CA = XX^b$

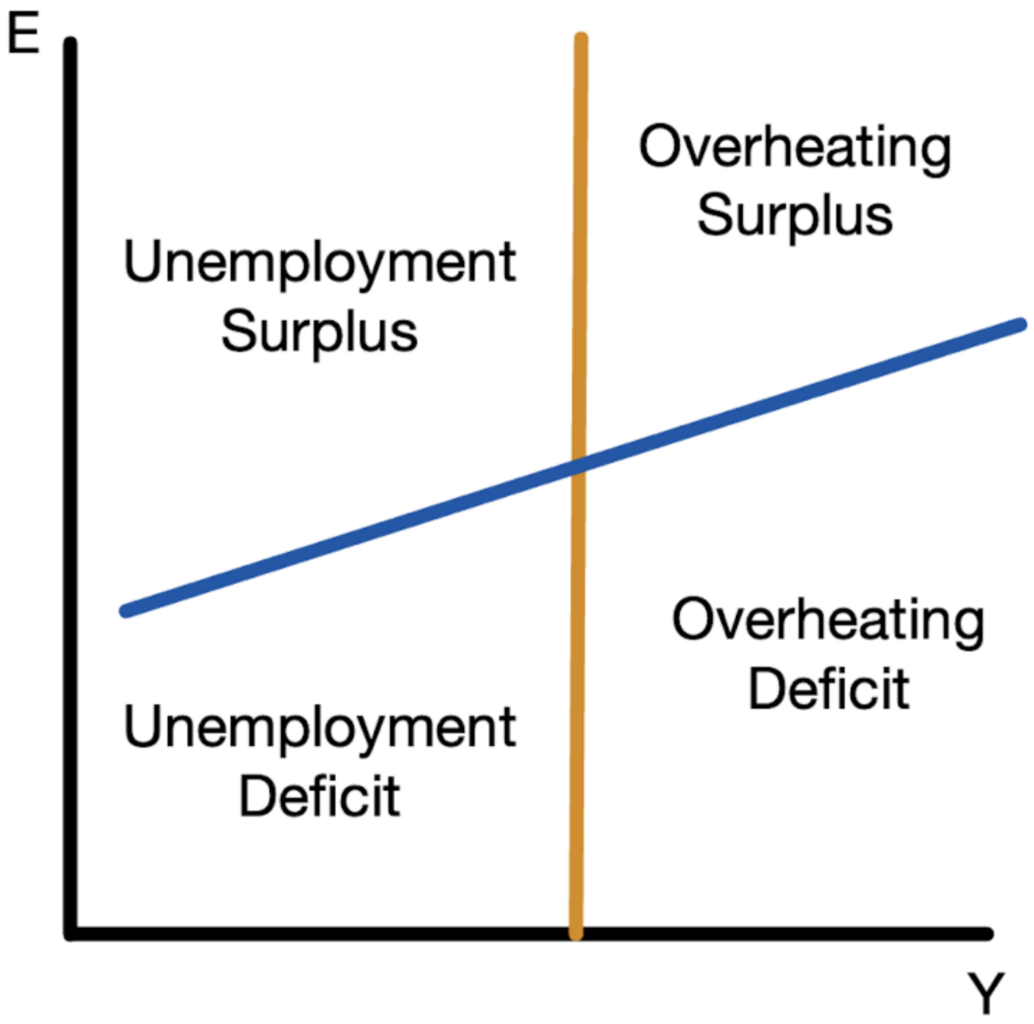


The XX curve is upward sloping



Four Zones of Discomfort

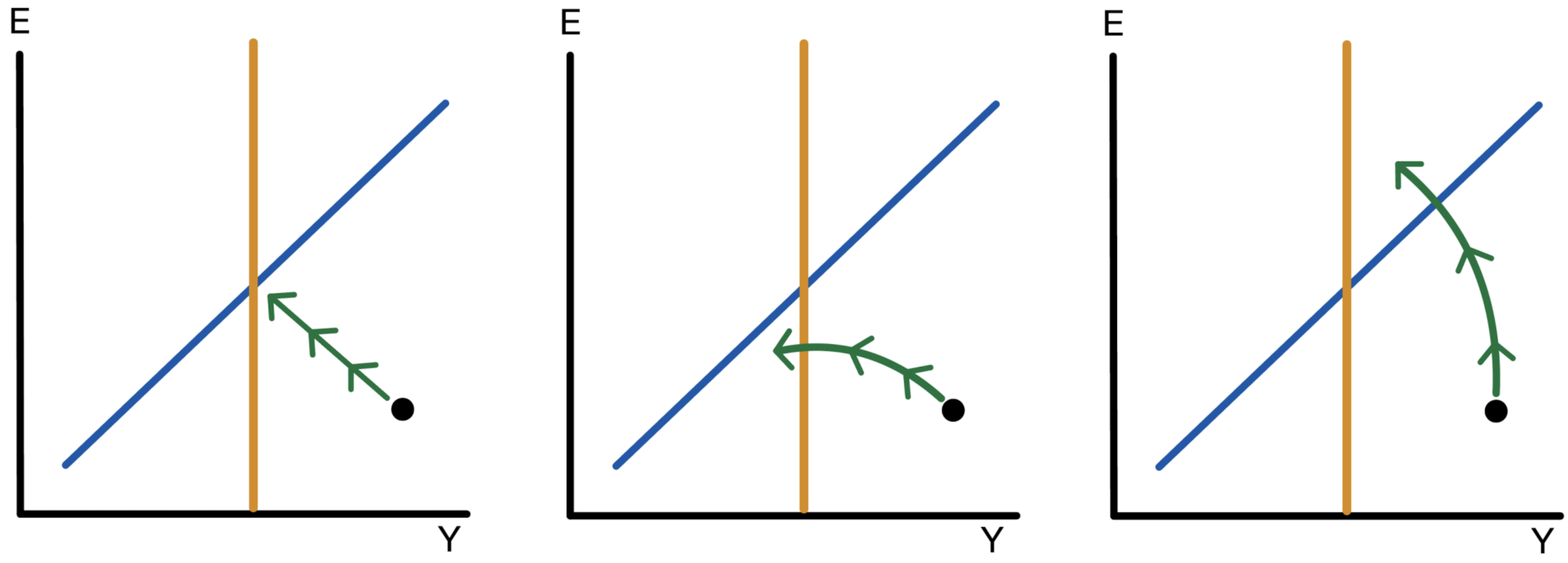
- Lack of internal balance:
  - Overheating when $Y > Y^f$
  - Unemployment when $Y < Y^f$
- Lack of external balance:
  - Deficit when $CA < XX^b$
  - Surplus when $CA > XX^b$
- Good tool to diagnose state of an economy



Dynamics

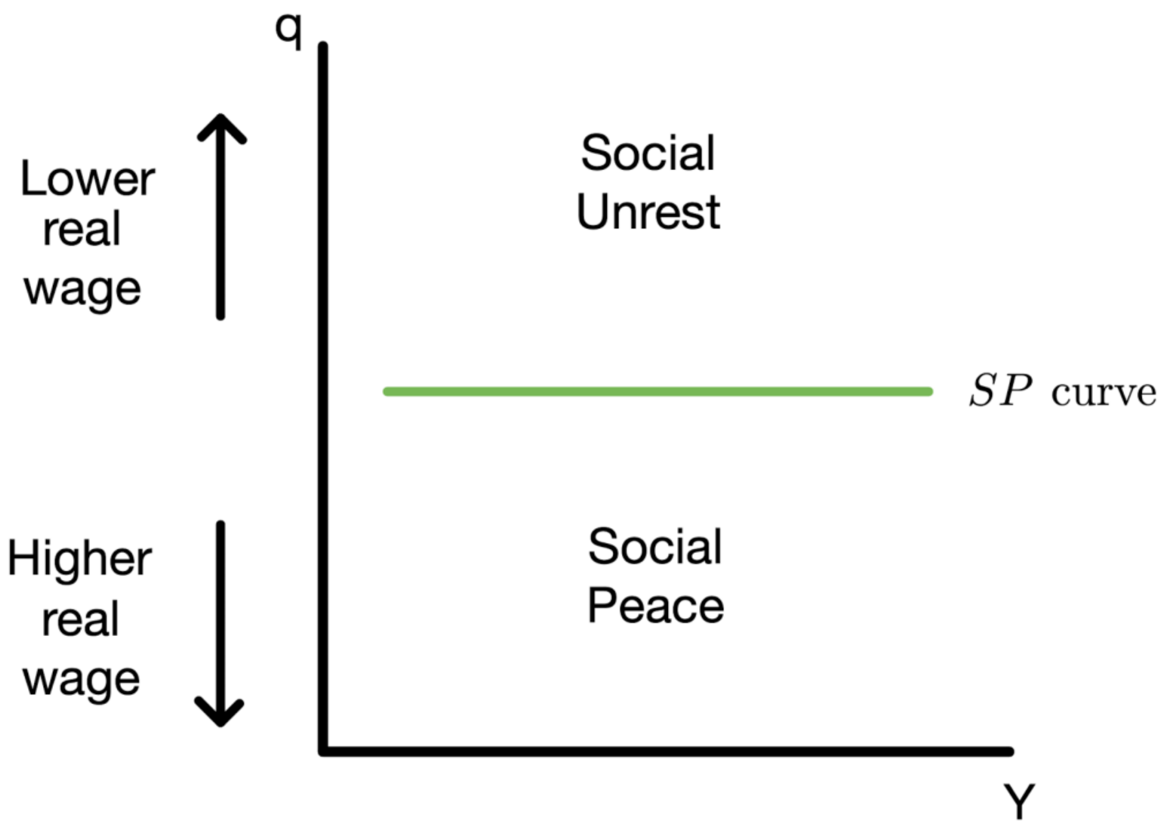
- Output adjusts to Y^f in the long run
- Current account adjusts to XX^b in the long run
- Otherwise, dynamics unspecified
- Can take a long time to return to balance
- Government policy can speed up return to balance

Possible Dynamics



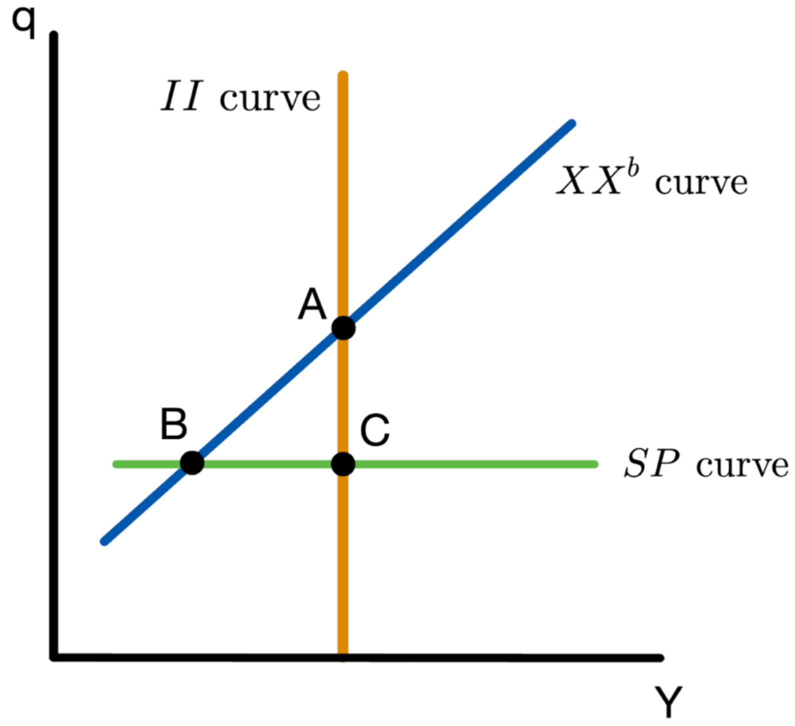
Social Peace

- When real wages are too low, there is **social unrest**
- Real depreciation associated with lower real wage



Economic and Social Objectives

- When SP curve is low enough, social and economic objectives are in conflict
- Assign evocative labels to A, B, C.



Point	Internal Balance	External Balance	Social Peace	Name
A	✓	✓	Conflict	
B	Unemployment	✓	✓	
C	✓	Deficit	✓	

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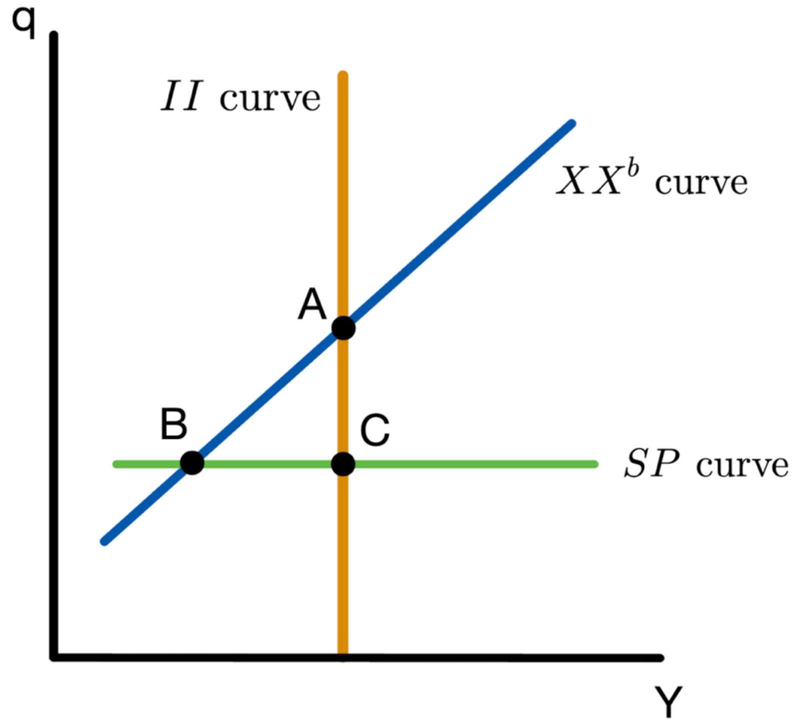
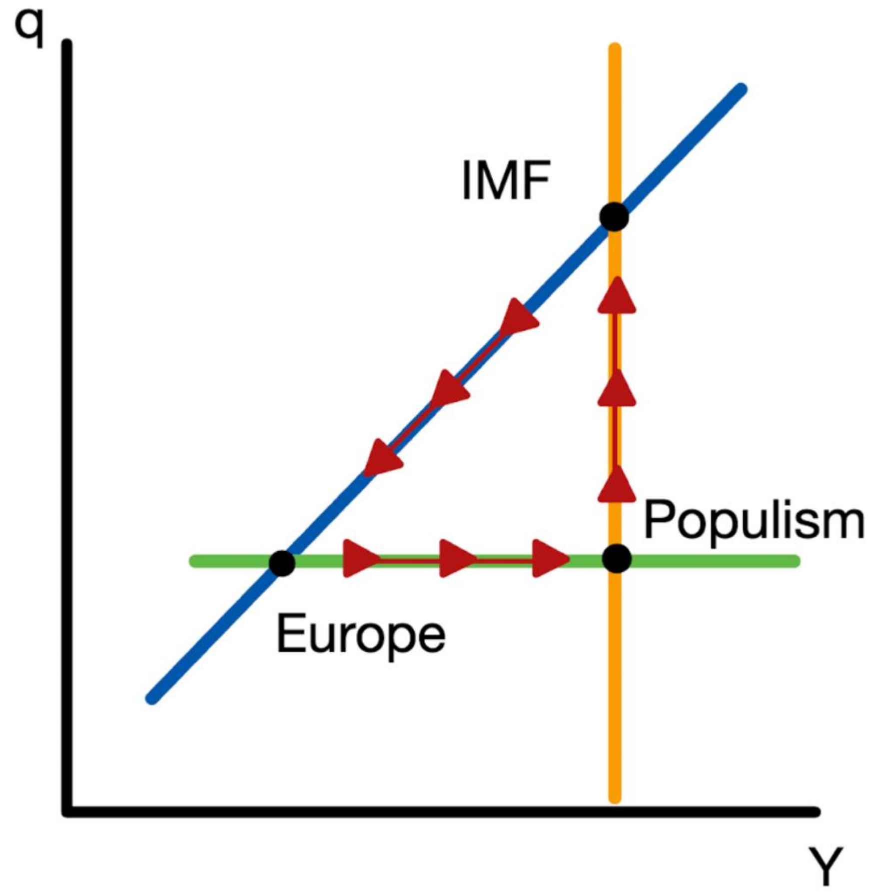


Point	Internal Balance	External Balance	Social Peace	Name
A	✓	✓	Conflict	IMF
B	Unemployment	✓	✓	Europe
C	✓	Deficit	✓	Populism

The Latin Triangle

- Economy cycles between IMF, Europe, Populism
- Describes dynamics in Latin America *scarily* well
- Economic cycles and how society sees itself
- No easy solutions



The II-XX with Domestic Demand and Real Exchange Rates

- Assume specific adjustment mechanisms to pin down dynamics
- Adjust to internal balance through the price level P
- Adjust to external balance through consumption C
- Study environmental sustainability (together with social peace and economic balance)

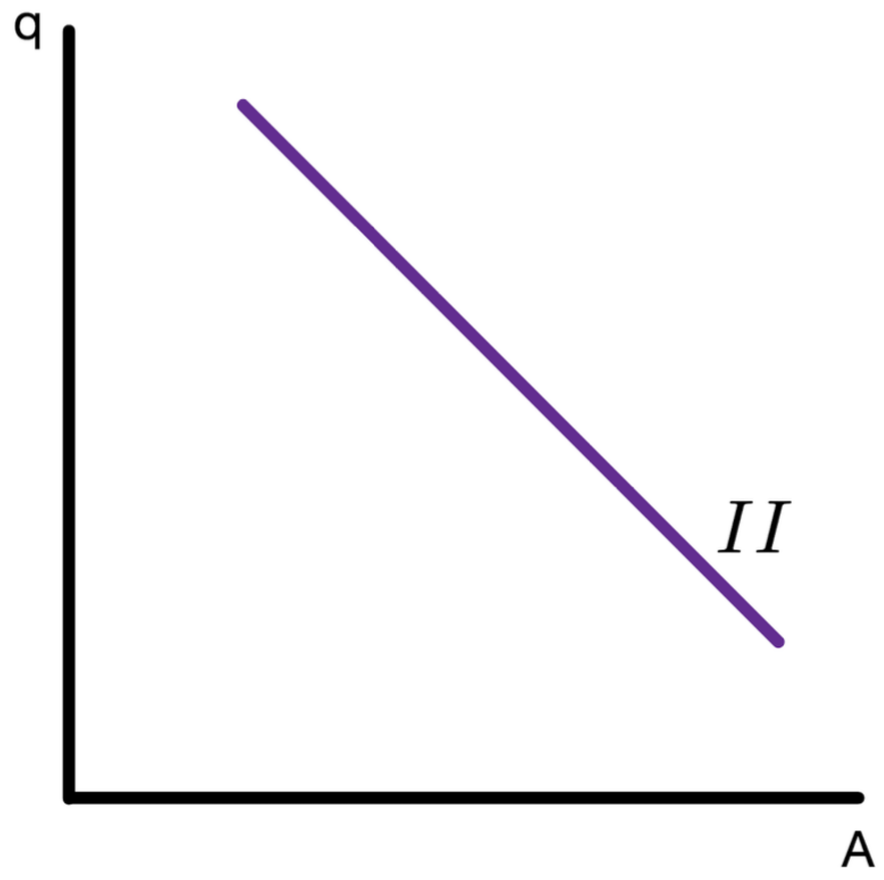
The II Curve

- Domestic demand

$$A = C + I + G$$
$$= C(Y - T) + I + G$$

- Demand for domestic goods

$$Z = C + I + G + CA$$
$$= C(Y - T) + I + G + CA(q, Y, Y^*)$$



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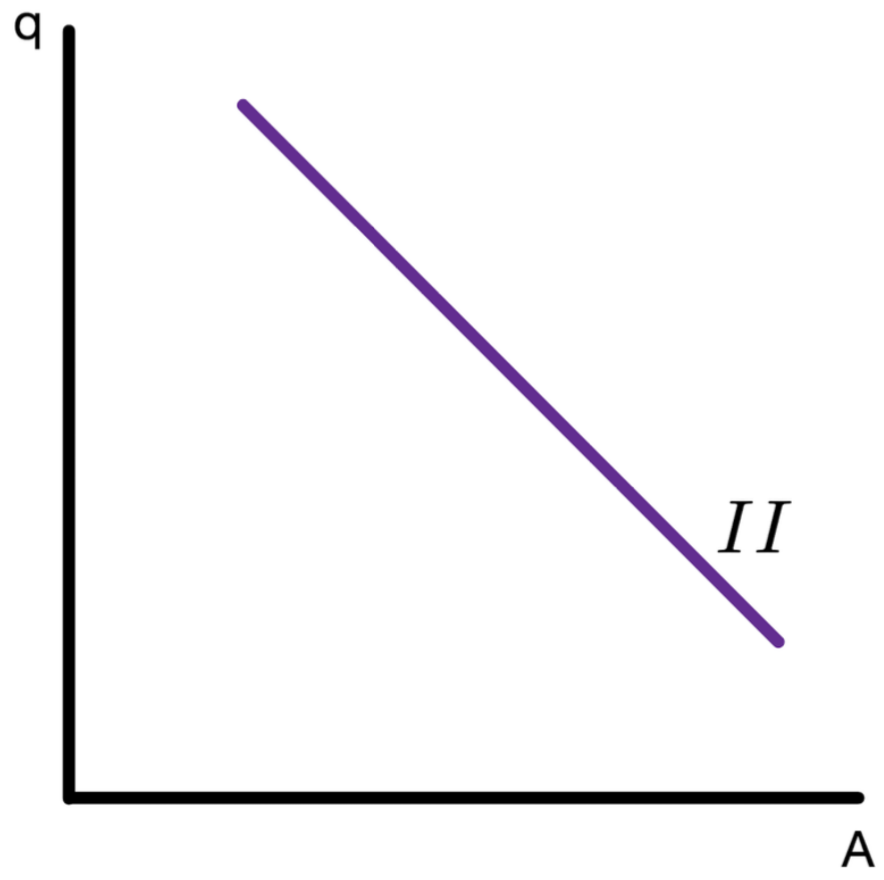
- Combine both

$$A = Y - CA(q, Y, Y^*)$$



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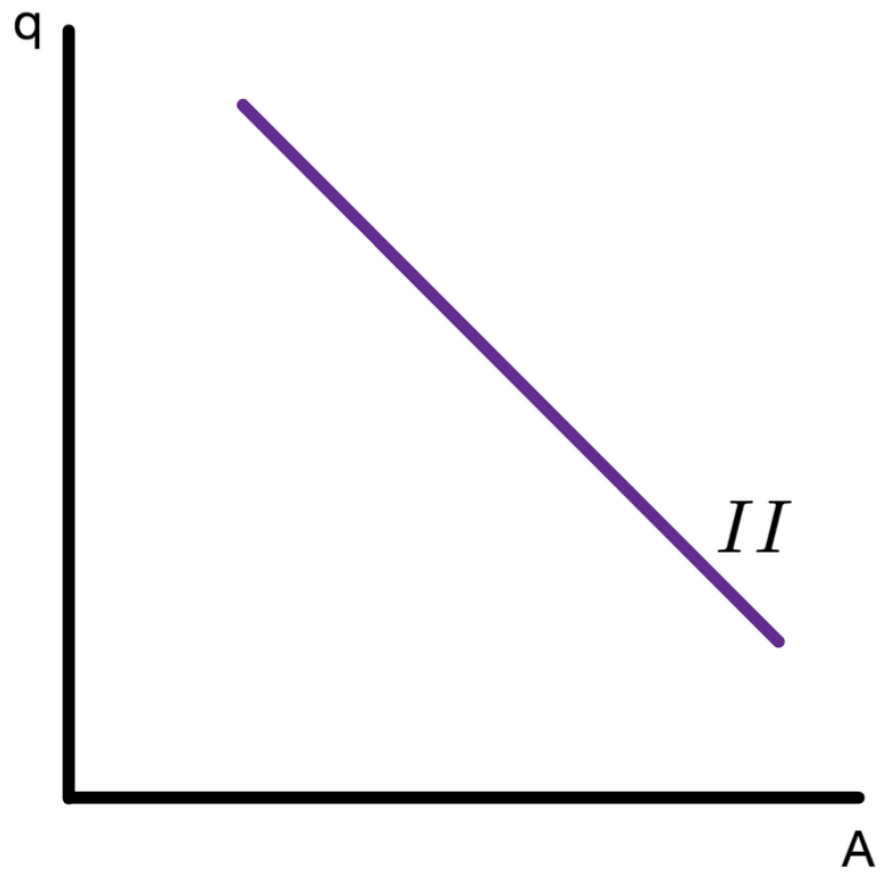
- Combine both

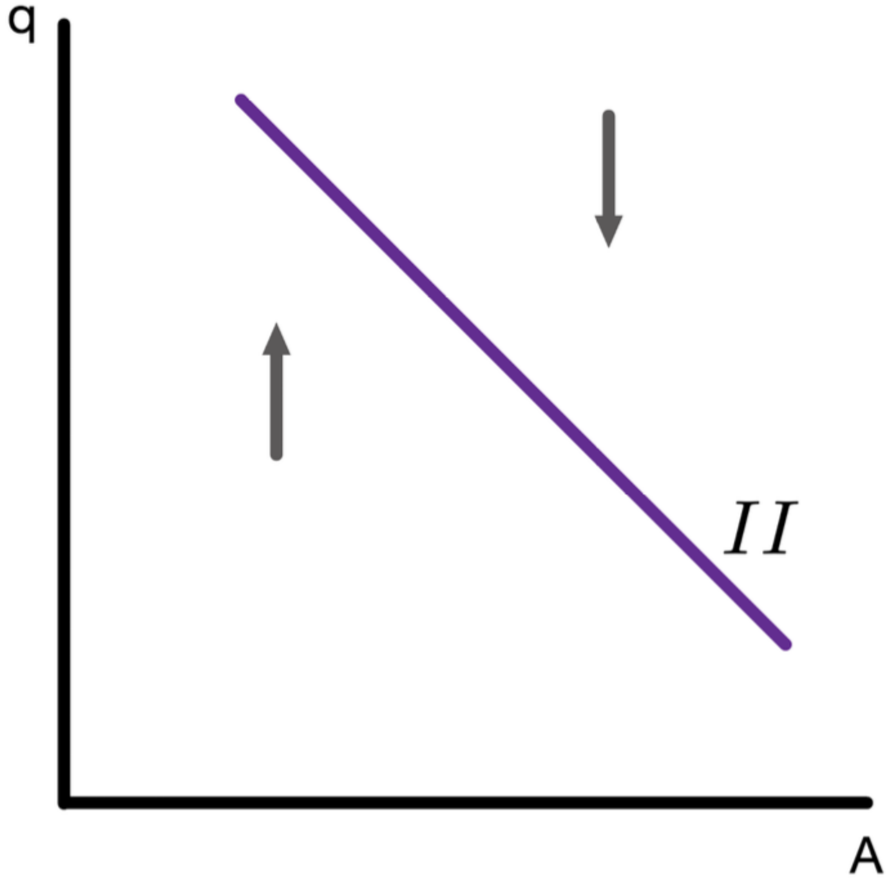
$$A = Y - CA(q, Y, Y^*)$$

- Set $Y = Y^f$

$$A = Y^f - CA(q, Y^f, Y^*)$$



Adjustment Dynamics for II Curve

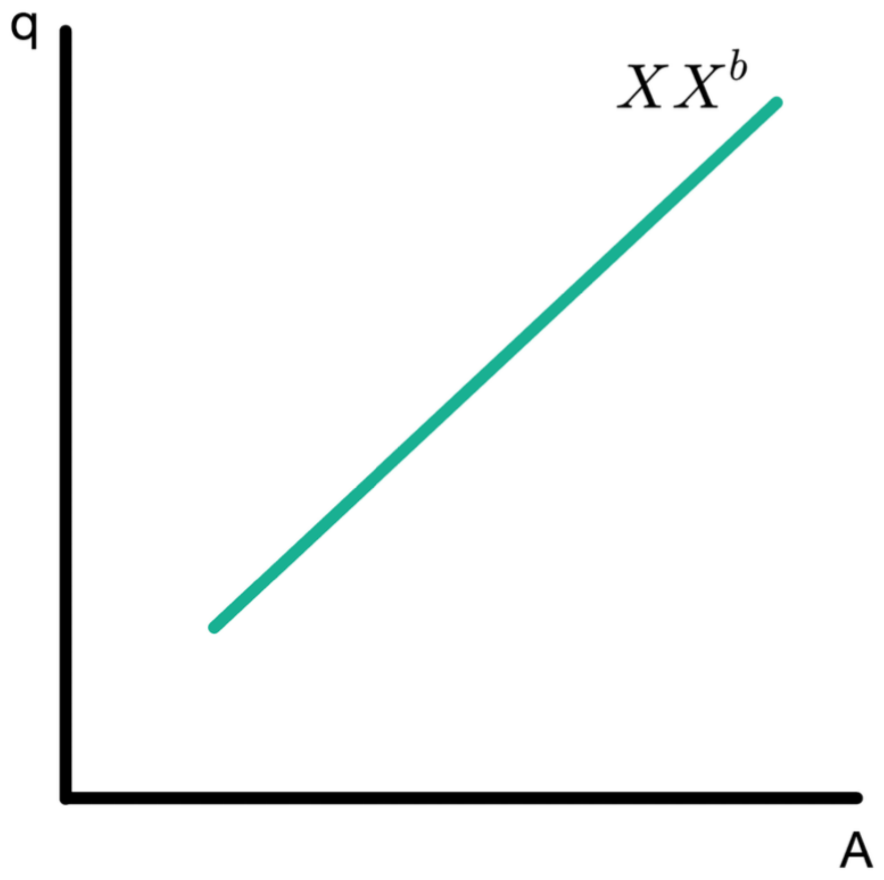


- The economy tends toward internal balance through adjustments in the price level P
  - When $Y > Y^f$, P is increasing
  - When $Y > Y^f$, P is decreasing
- Changes in P make Y approach Y^f

The XX Curve

Start with domestic demand

$$A = C(Y - T) + I + G$$



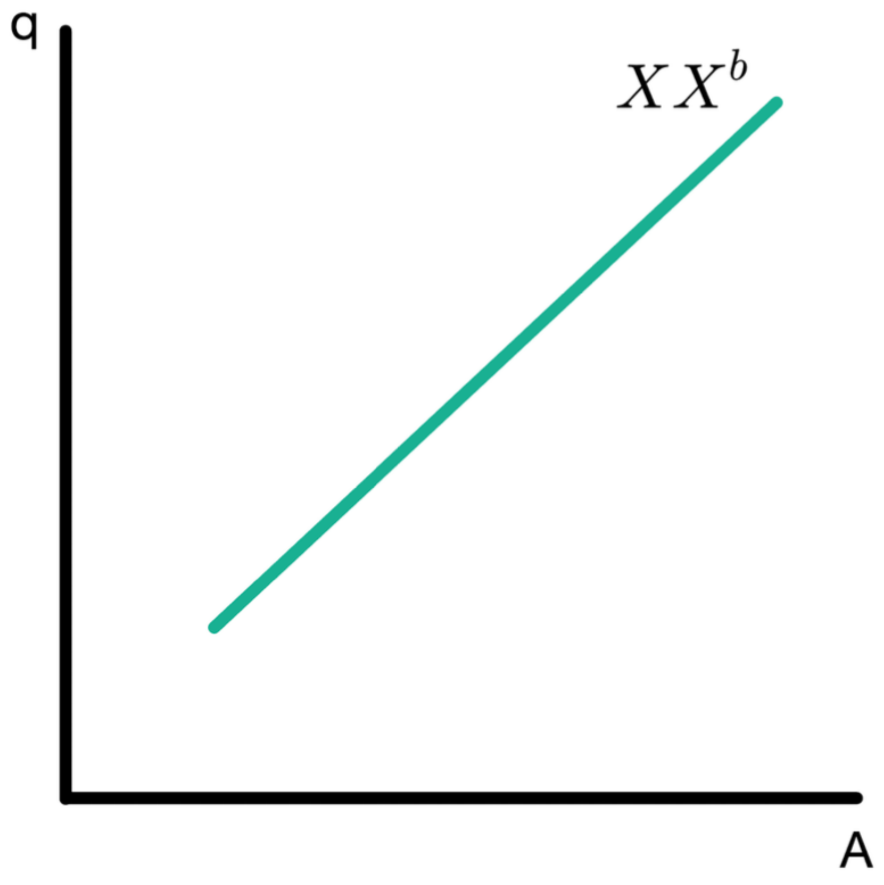
The XX Curve

Start with domestic demand

$$A = C(Y - T) + I + G$$

Solve for Y

$$Y = C^{-1}(A - I - G) + T$$



The XX Curve

Start with domestic demand

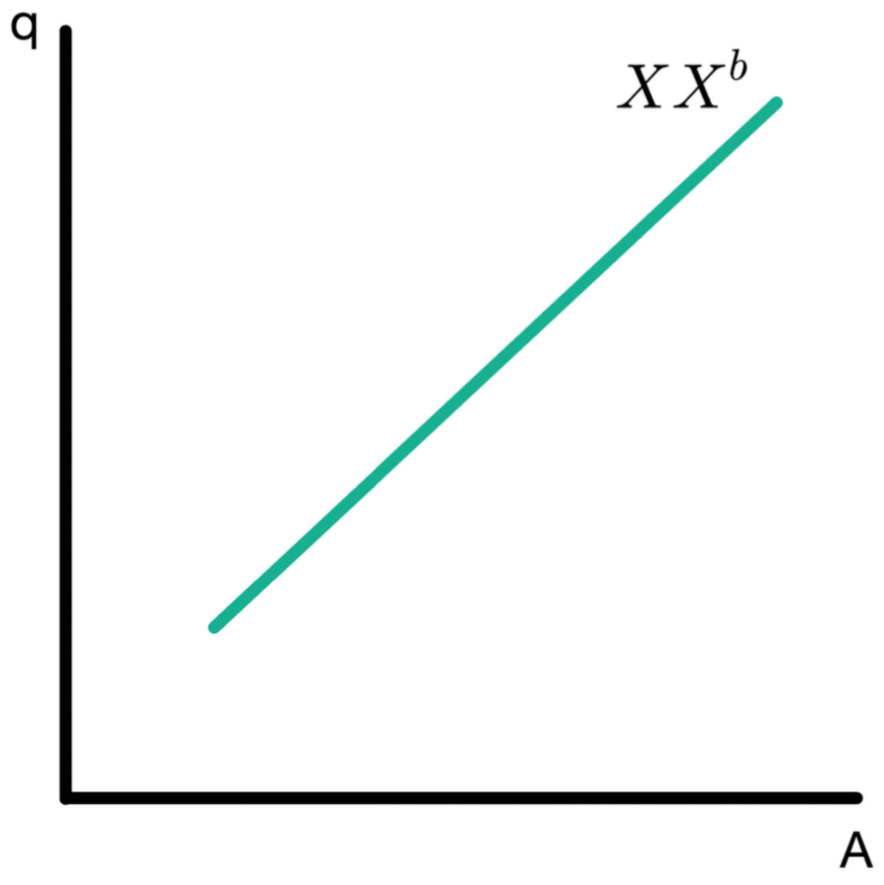
$$A = C(Y - T) + I + G$$

Solve for Y

$$Y = C^{-1}(A - I - G) + T$$

Insert in

$$\begin{aligned} XX^b &= CA(q, Y, Y^*) \\ &= CA(q, C^{-1}(A - I - G) + T, Y^*) \end{aligned}$$



The XX Curve

Start with domestic demand

$$A = C(Y - T) + I + G$$

Solve for Y

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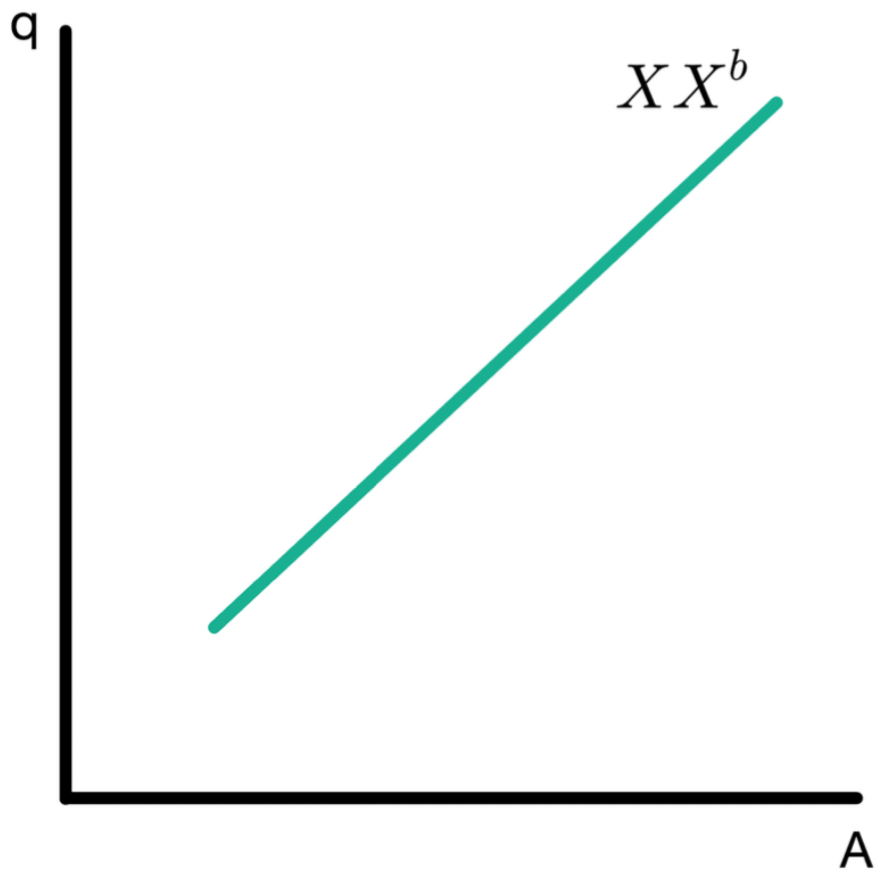
Insert in

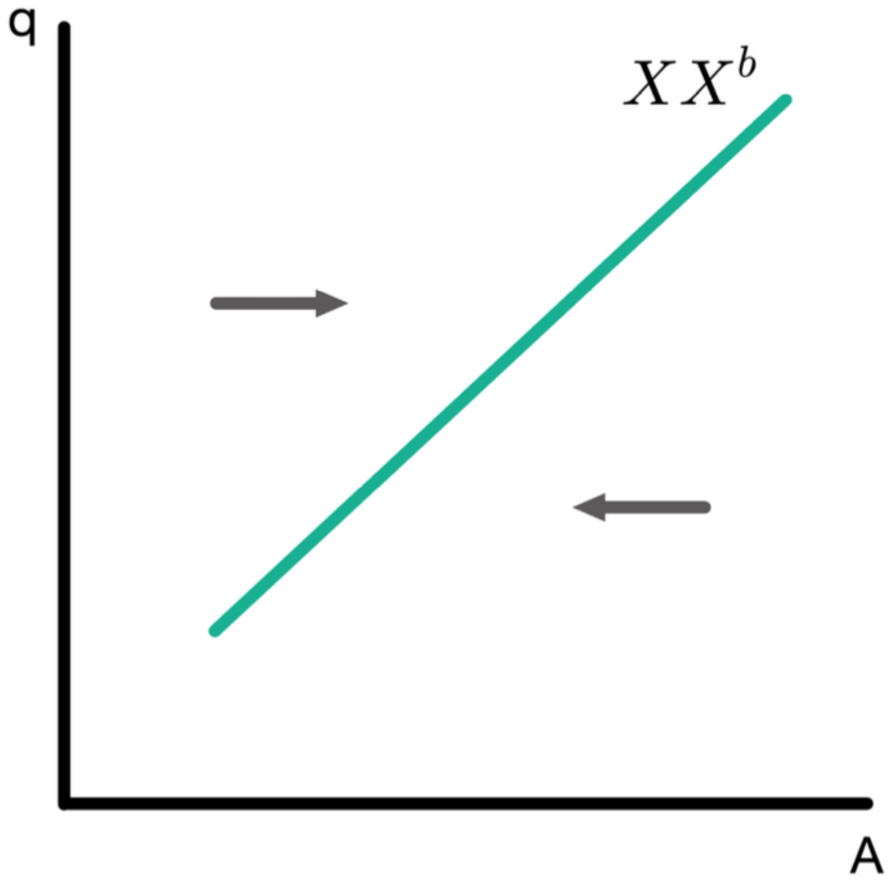
$$\begin{aligned} XX^b &= CA(q, Y, Y^*) \\ &= CA(q, C^{-1}(A - I - G) + T, Y^*) \\ &= \widetilde{CA}(q, A, Y^*) \end{aligned}$$

where the function $\widetilde{CA}$ is defined by

$$\widetilde{CA}(q, A, Y^*) \equiv CA(q, C^{-1}(A - I - G) + T, Y^*)$$

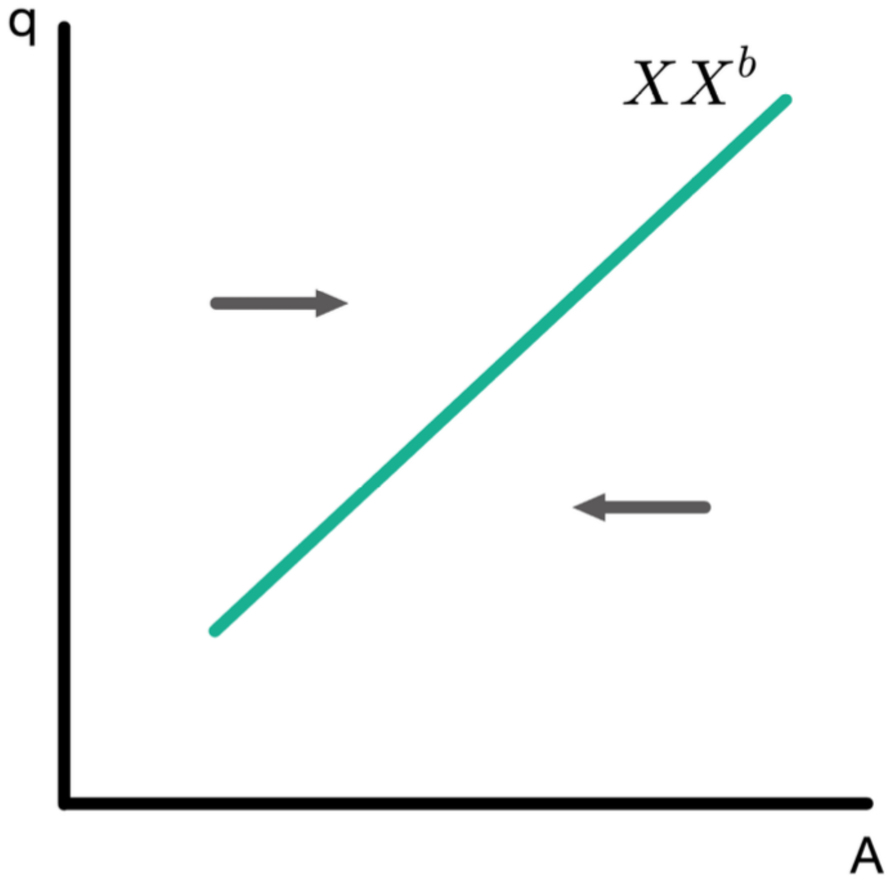


Adjustment Dynamics for XX Curve



Two sources of adjustment:

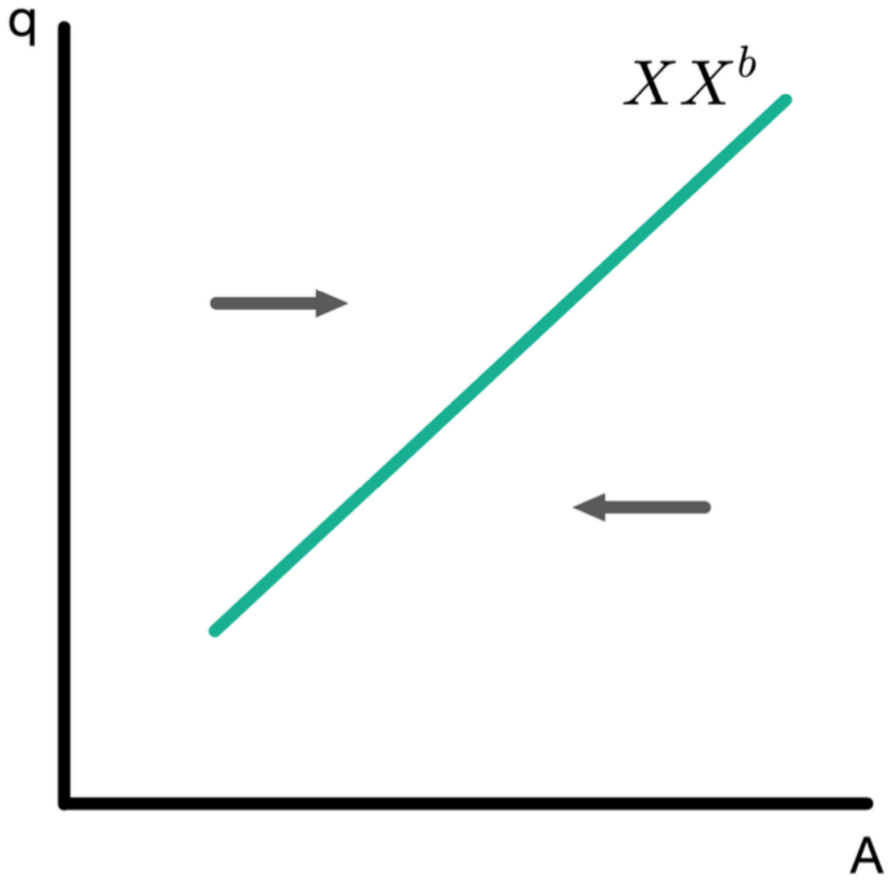
Adjustment Dynamics for XX Curve



Two sources of adjustment:

- An aggregate budget constraint: we cannot finance too high a level of consumption with ever-increasing borrowing.

Adjustment Dynamics for XX Curve



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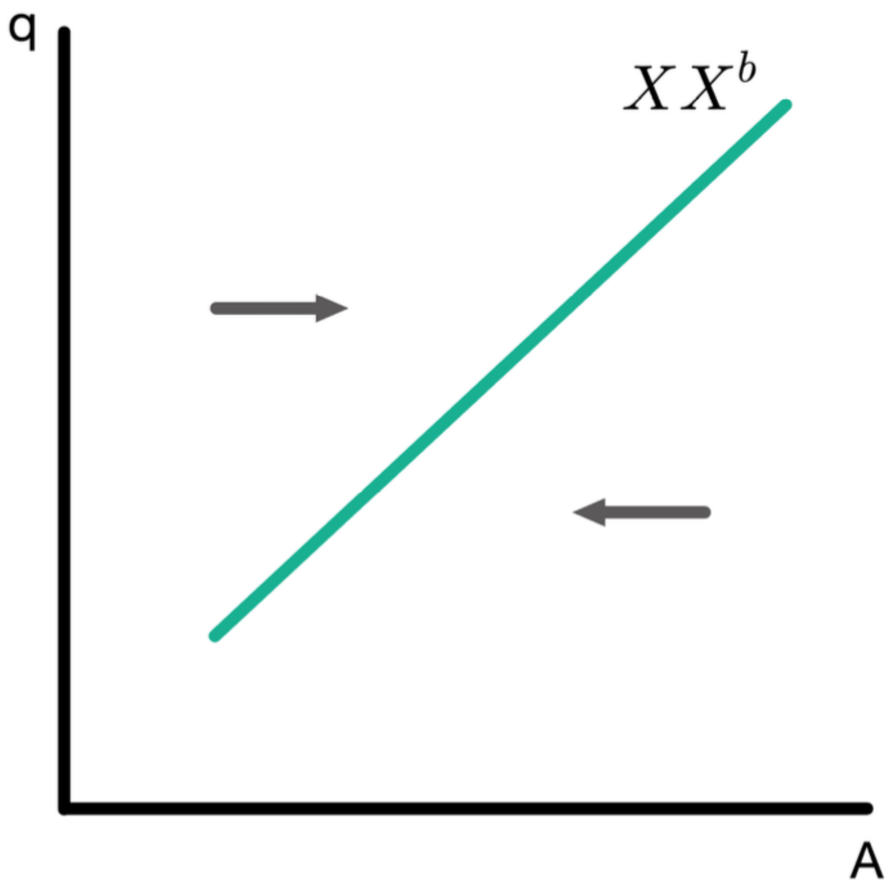
  - When $CA < XX^b$, consumption is decreasing over time through budget constraint provide

Adjustment Dynamics for XX Curve



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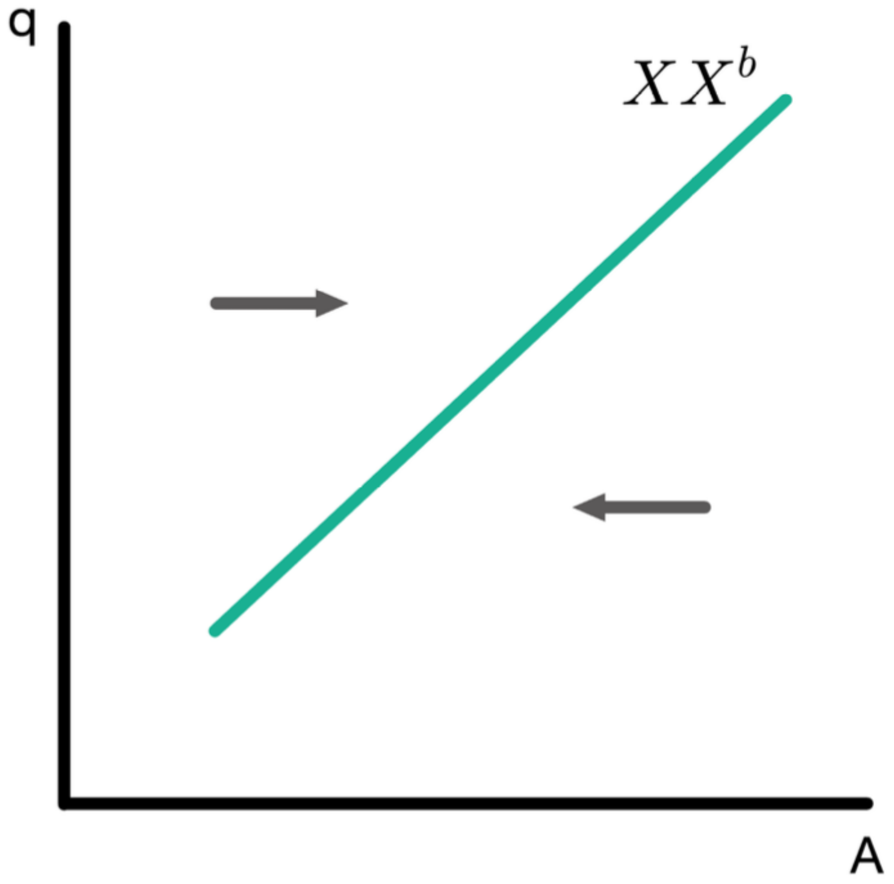
- We value consumption and will not choose to consume less in every period if it can consume more (nonsatiation)

Adjustment Dynamics for XX Curve



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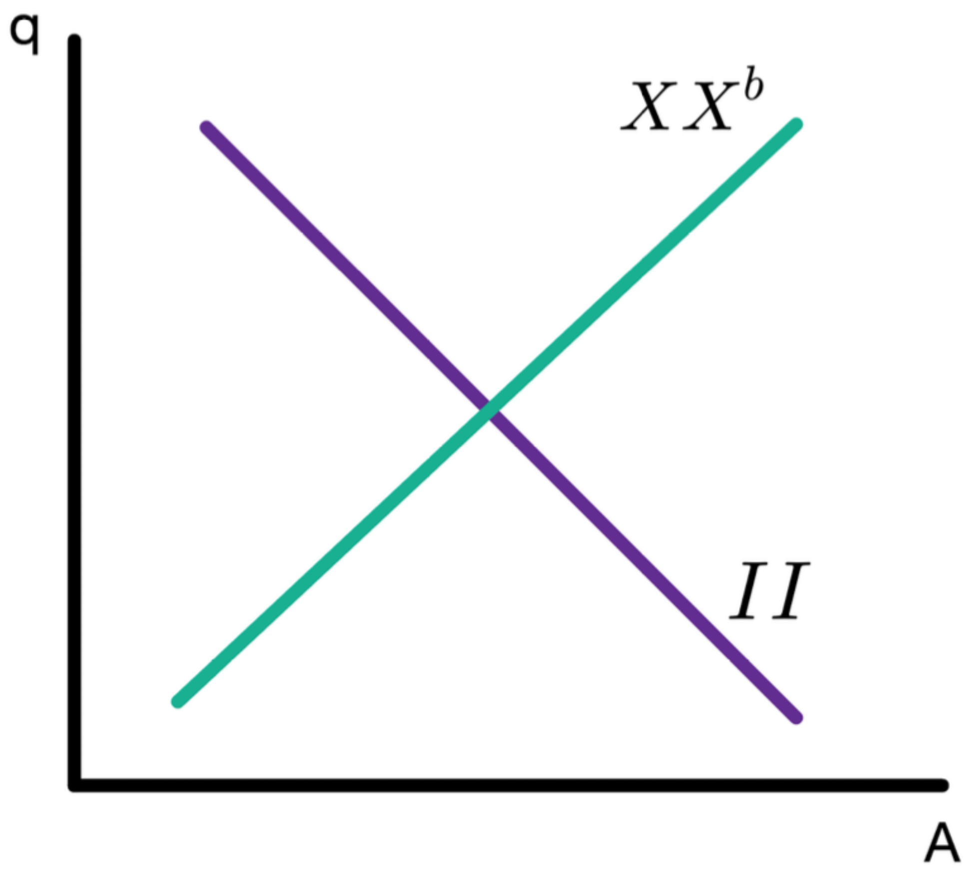
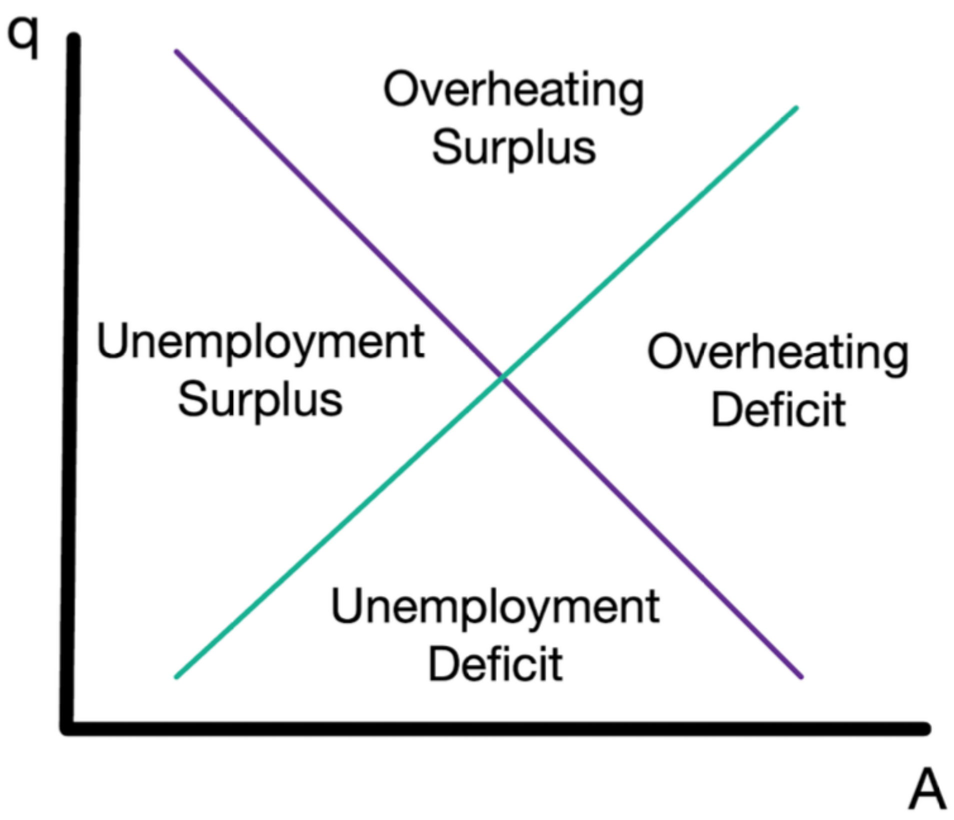
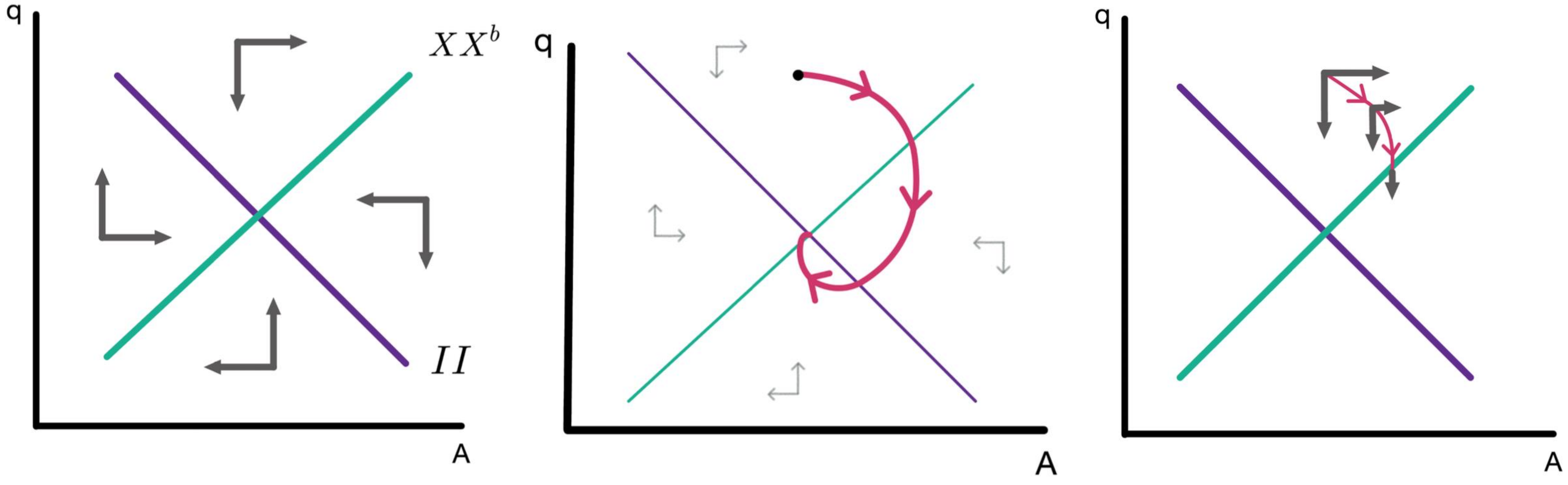
  - When $CA < XX^b$, consumption is decreasing over time through budget constraint provide
- We value consumption and will not choose to consume less in every period if it can consume more (nonsatiation)
  - When $CA > XX^b$, consumption is increasing over time through nonsatiation

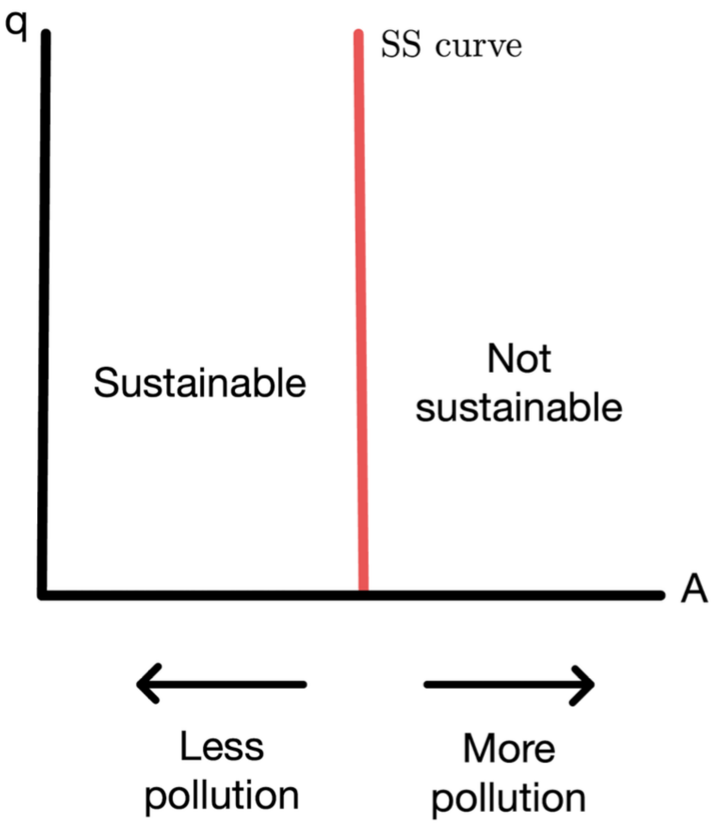
Four Zones of Economic Discomfort

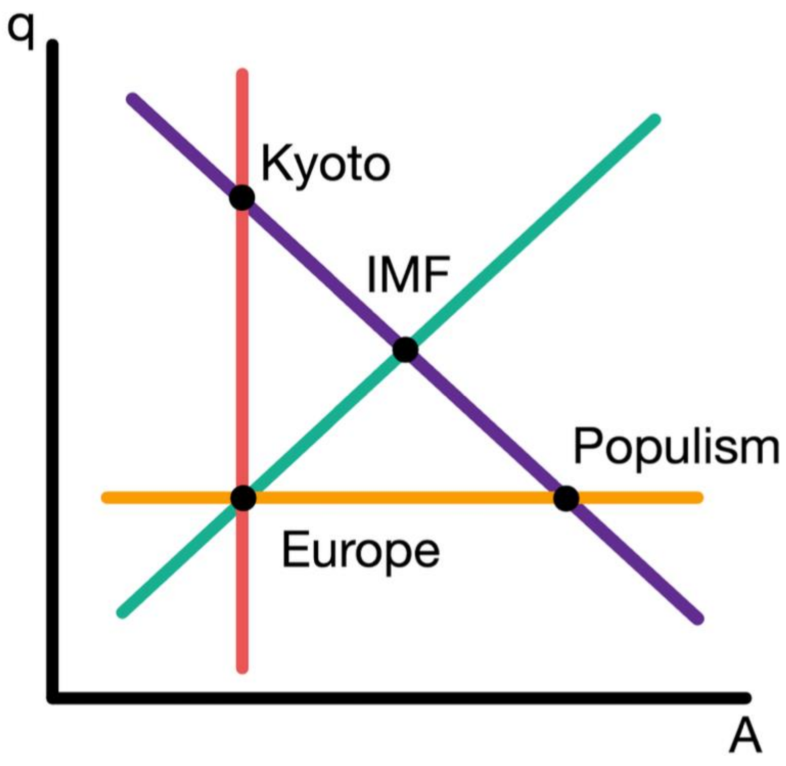




Dynamics for II and XX Combined



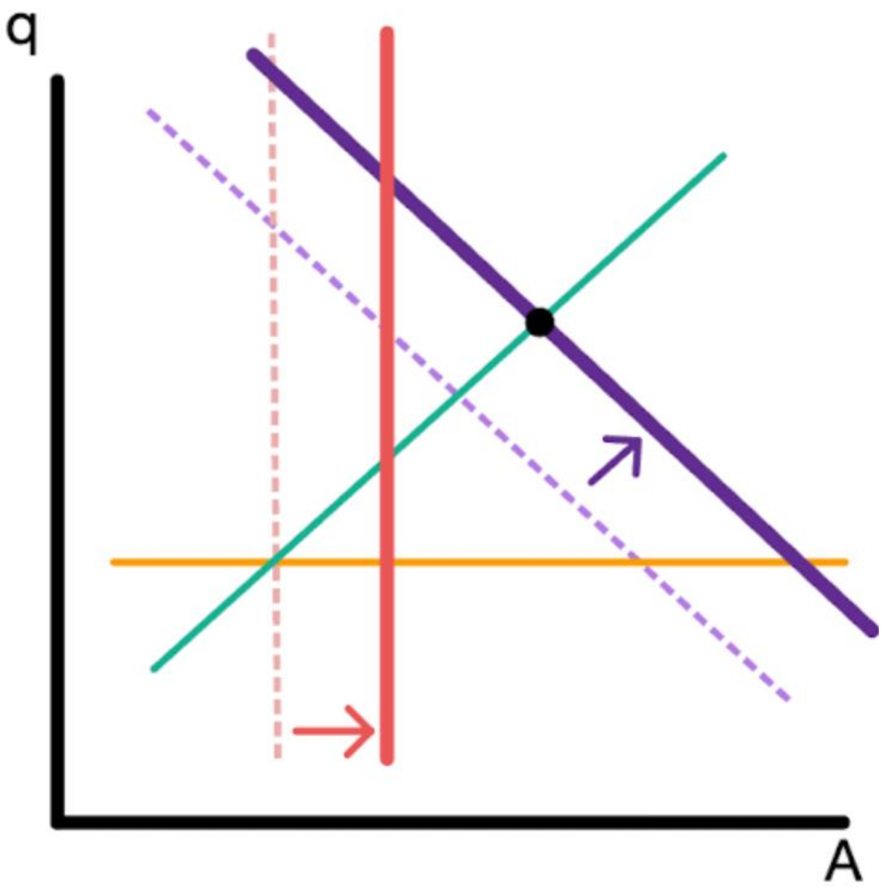




	Internal Balance	External Balance	Social Peace	Environment
IMF	✓	✓	Conflict	Pollution
Europe	Unemployment	✓	✓	✓
Populism	✓	Deficit	✓	High Pollution
Kyoto	✓	Surplus	Conflict	✓

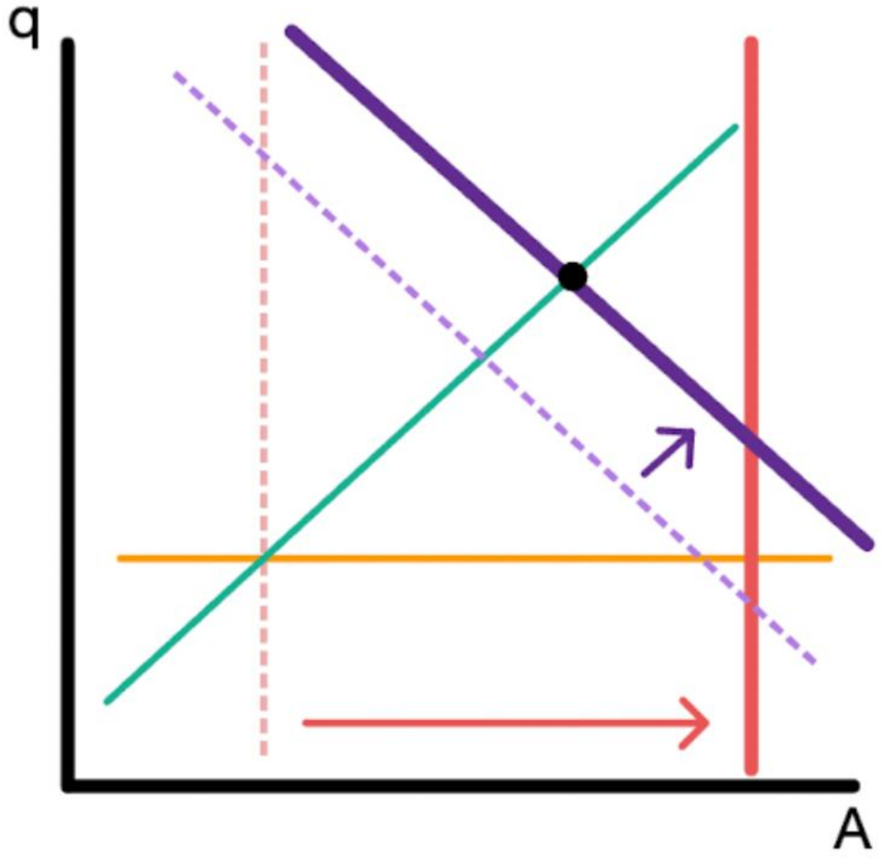
Can Technological Progress Help?

- It depends!
- Technological progress: A in the production function $Y = AK^{1-\alpha} N^{\alpha}$
  - Can help us produce more without harming the environment (SS shifts to the right)
  - But also makes Y^f go up (II shifts to the right)
- If II shifts a lot and SS shifts little, then environmental sustainability still incompatible with internal and external balance



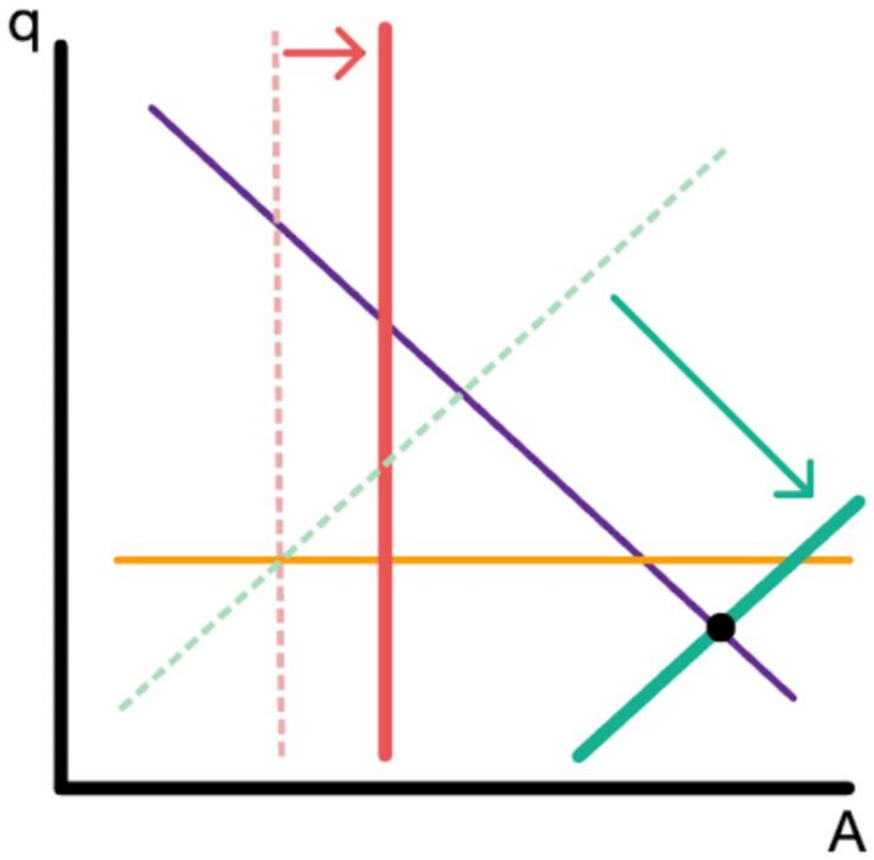
Can Technological Progress Help?

- If *II* shifts a little compared to *SS*, then environmental sustainability can be achieved together with internal and external balance!



Can Technological Progress Help?

- Technology can also shift the XX
  - For example, higher productivity abroad can increase Y^*
- Technology that make a larger CA deficit sustainable shift the XX down
- Large XX shifts and small SS shifts do not help solve the conflict between environmental and economic objectives



Can Technological Progress Help?

- Shifts in XX that are relative small compared to shifts in SS can solve the conflict between environmental and economic objectives!